Fallacies :- (Invalid arguments).

↳ They are errors in reasoning that lead to invalid arguments.
↳ They may seem convincing but don't hold upto scrutiny.

(1.) Affirming the Consequent / Fallacy of assuming converse :-

$$\left.\begin{array}{l} p \to q \\ q \\ \hline \therefore p \end{array}\right\} \quad \begin{array}{l} \text{no.} \\ \text{Invalid argument.} \end{array}$$

↳ $((p \to q) \wedge q) \to p$ is a tautology.

p	q	$p \to q$ (x)	$x \wedge q$ (y)	$y \to p$
T	T	T	T	T
F	T	T	T	F

not a tautology.

eg. P_1 : If you have the flu (p), then you have a fever (q).
P_2 : You have a fever. q

$\therefore Q$: You have the flu. p

$$\left.\begin{array}{l} p \to q \\ q \\ \hline \therefore p \end{array}\right\} \quad \text{Invalid.} \ \times$$

2. Denying the antecedent / Fallacy of assuming inverse :-

$p \rightarrow q$

$\boxed{\neg p}$

$\therefore \neg q$

} Invalid.

p	q	$p \rightarrow q$ (x)	$x \wedge \neg p$ (y)	$y \rightarrow \neg q$
F	T	T	T	F
F	F	T	T	T

$\times$ not a tautology.

eg. P_1 : If it is a bird, then it can fly.

P_2 : It is not a bird.

$\therefore Q$: Therefore, it cannot fly.

} Invalid.

③ Non sequitur :-

↳ There's a gap in the logic that isn't explained.

$$\begin{array}{l} p \\ q \\ r \\ \hline \therefore s \end{array} \Bigg\} \text{ Invalid.}$$

eg.

P_1 : Today is sunny.
P_2 : Yesterday was rainy.
P_3 : If you study hard, then you might pass.

Q : All birds can fly.

} Invalid.

1. I. p : The election is decided.
q : The votes have been counted.
Give the english sentence of these compound propositions.
a) $\neg q \rightarrow \neg p$: If the votes haven't been counted, then the election is not decided.
b) $p \leftrightarrow q$: p if and only if q.
c) $\underbrace{\neg p} \vee \underbrace{(\neg p \wedge q)}$: Either the election is not decided, or else the election is not decided and the votes have been counted.

II. p : You have the flu.
q : You miss the final examination.
r : You pass the course.
a) $\underbrace{(p \rightarrow \neg r)} \vee \underbrace{(q \rightarrow \neg r)}$
b) $\underbrace{(p \wedge q)} \vee \underbrace{(\neg q \wedge r)}$.

a) If you have the flu, then you won't pass the course, or If you miss the final examination, then you won't pass the course.

b) Either you have the flu and you miss the final examination, or else you do not miss the final exam and do pass the course.

(2.) State the converse, contrapositive & inverse of each of them :-

a) I go to the beach whenever it is a sunny summer day.
(the beach → q, whenever boxed, "it is a sunny summer day" → p)

b) When I stay up late, it is necessary that I sleep until noon. (q)
(I stay up late → p)

$\boxed{p \to q}$, Converse : $q \to p$, Inverse : $\neg p \to \neg q$, Contra : $\neg q \to \neg p$.

Sol. a) Converse : If I go to the beach then it's a sunny summer day.
Contrapositive : If I do not go to the beach then it's not a sunny summer day.
Inverse : If it's not a sunny summer day then I will not go to the beach.

$$\boxed{p \to q \equiv q \text{ whenever } p}$$

b) Converse : If I sleep until noon, then I stay up late.
Contrapositive : If I do not sleep until noon, then I did not stay up late.
Inverse : If I did not stay up late, then I don't sleep until noon.

$\boxed{\text{When}}$ p, $\boxed{\text{it}}$ is necessary q

q is necessary for p

= "$\underbrace{\text{I sleep until noon}}_{q}$" $\underbrace{\text{is necessary for}}$ "$\underbrace{\text{Staying up late}}$".

$\boxed{\text{When}}$ "$\underbrace{\text{I stay up late}}_{p}$", $\underbrace{\text{it is necessary that}}_{\hookrightarrow q}$ (p

"I sleep until noon".

(3.) Which one of them is a tautology, contradiction & contingency.

a) $p \to \neg p$

b) $p \oplus (p \vee q)$

c) $p \leftrightarrow \neg p$

d) $(p \wedge q) \to (p \vee q)$

e) $(q \to \neg p) \leftrightarrow (p \leftrightarrow q)$

f) $(p \leftrightarrow q) \oplus (p \leftrightarrow \neg q)$

$p \to \neg p$

Sol. a)

p	$\neg p$	$p \to \neg p$
T	F	F
F	T	T

contingency.

b) $p \oplus (p \vee q)$

p	q	$p \vee q$ (x)	$p \oplus x$
T	T	T	F
T	F	T	F
F	T	T	T
F	F	F	F

contingency.

c) $p \leftrightarrow \neg p$

p	$\neg p$	$p \leftrightarrow \neg p$
T	F	F
F	T	F

Contradiction.

d) $\overbrace{(p \wedge q)}^{P} \rightarrow \overbrace{(p \vee q)}^{Q} \equiv \neg(p \wedge q) \vee (p \vee q)$

$\equiv (\neg p \vee \neg q) \vee (p \vee q)$

$\equiv (\neg p \vee p) \vee (\neg q \vee q) \equiv T$ Tautology.

e) $(q \rightarrow \neg p) \leftrightarrow (p \leftrightarrow q)$

p	q	$\neg p$	$q \rightarrow \neg p$ (x)	$p \leftrightarrow q$ (y)	$x \leftrightarrow y$
T	T	F	F	T	F
T	F	F	T	F	F
F	T	T	T	F	F
F	F	T	T	T	T

Contingency.

f) $(p \leftrightarrow q) \oplus (p \leftrightarrow \neg q)$

p	q	$\neg q$	$p \leftrightarrow q$ (x)	$p \leftrightarrow \neg q$ (y)	$x \oplus y$
T	T	F	T	F	T
T	F	T	F	T	T
F	T	F	F	T	T
F	F	T	T	F	T

tautology.